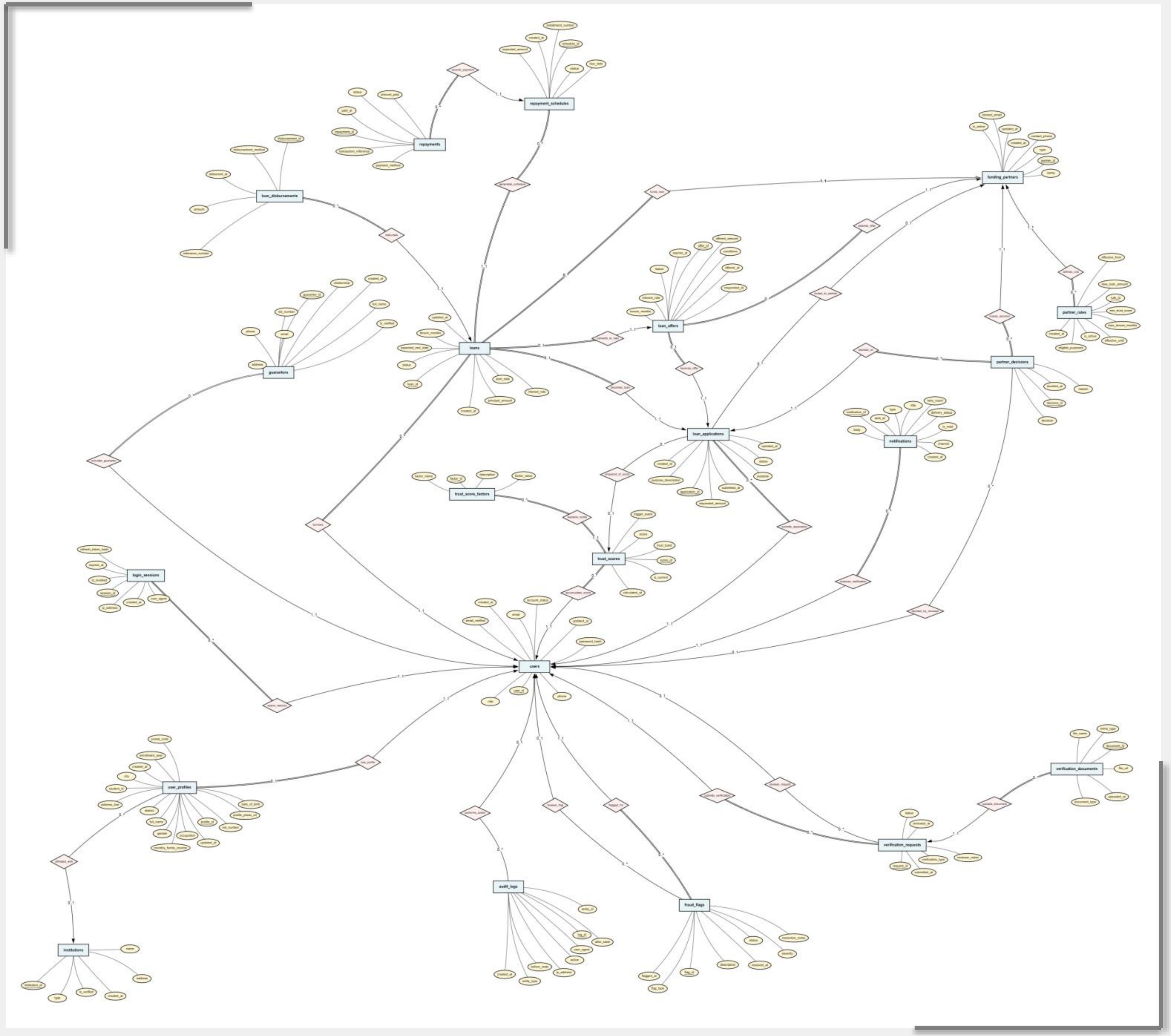
# ShohojRin
# ER Diagram

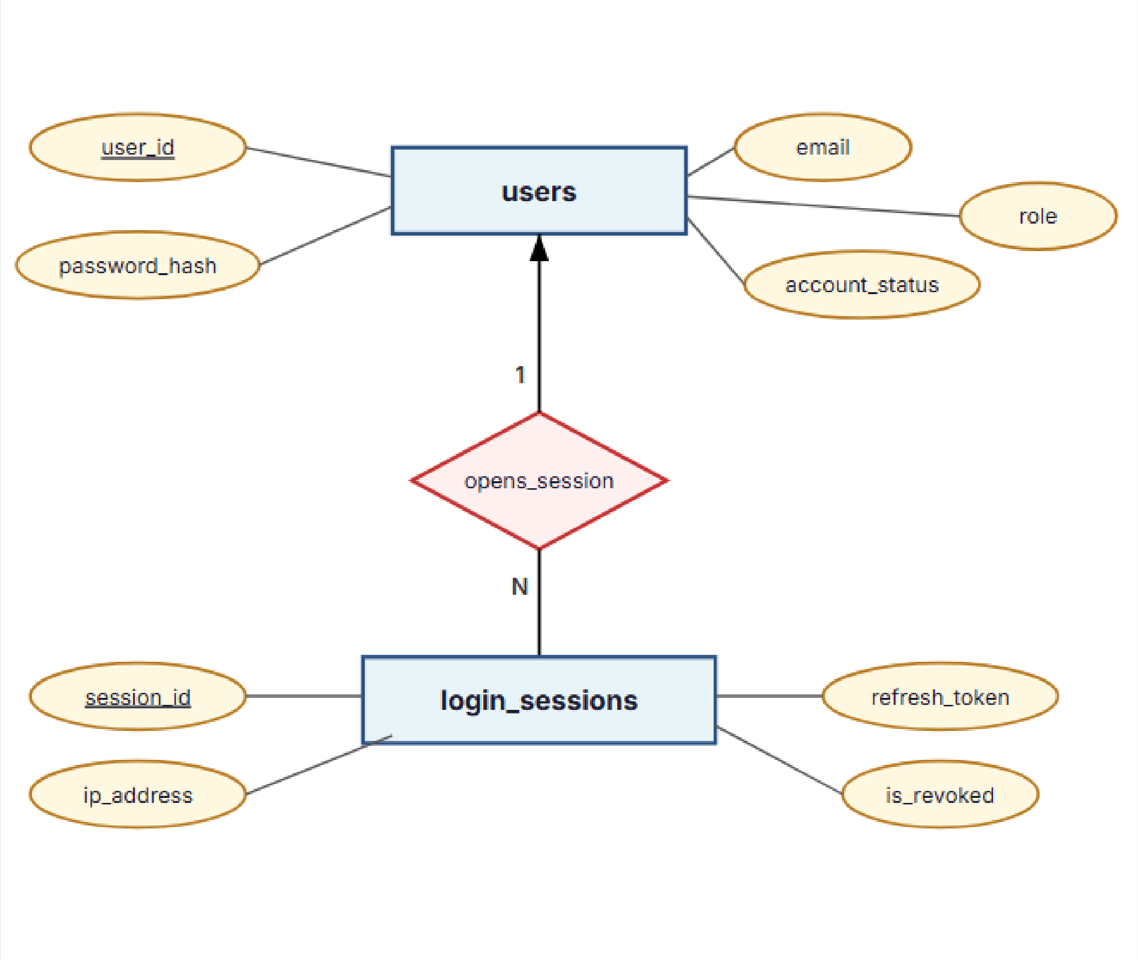
Trust-Based Verification & Loan Orchestration Platform |

- 21 Entities
- 29 Relationships
- Chen's Notation



# Module 1 — Authentication & User Management

Core identity, credentials, and session control




## Purpose

Manages authentication identity separately from profile data

Supports multi-device sessions with server-side revocation

Follows 3NF — credentials change independently from biographical data

## Entities

users, login_sessions

## Key Relationships

users → login_sessions (1:N) — one user opens many sessions

Sessions can be individually revoked without affecting the user record

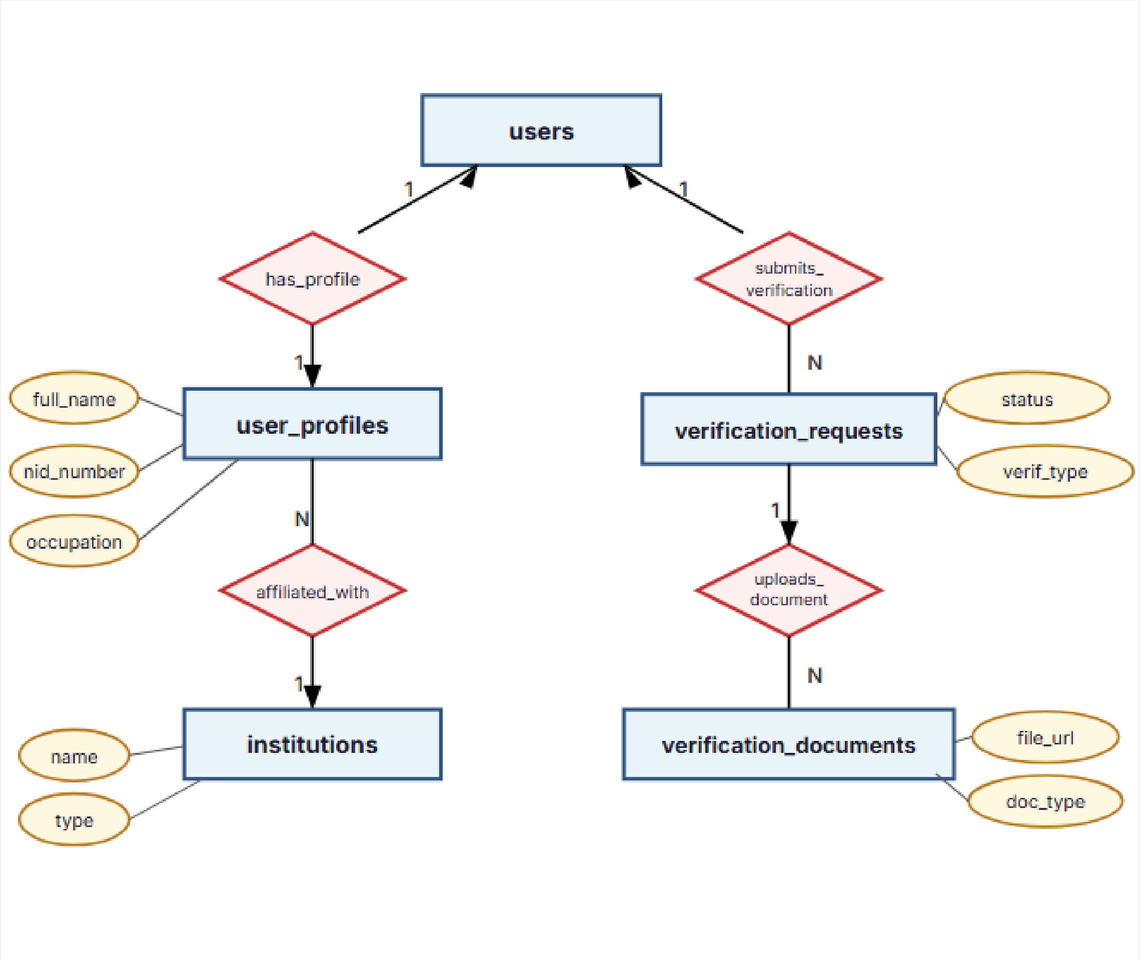
## Design Rationale

JWT alone cannot support forced logout; sessions enable revocation

User table is kept lean — only credentials and role

# Module 2 — User Profile & Verification

Profile data, institution normalization, document verification workflow




## Purpose

Separates profile data from authentication for normalization
Institutions are normalized — avoids repeating "BUET" across thousands of profiles
Verification is a workflow, not a boolean flag

## Entities

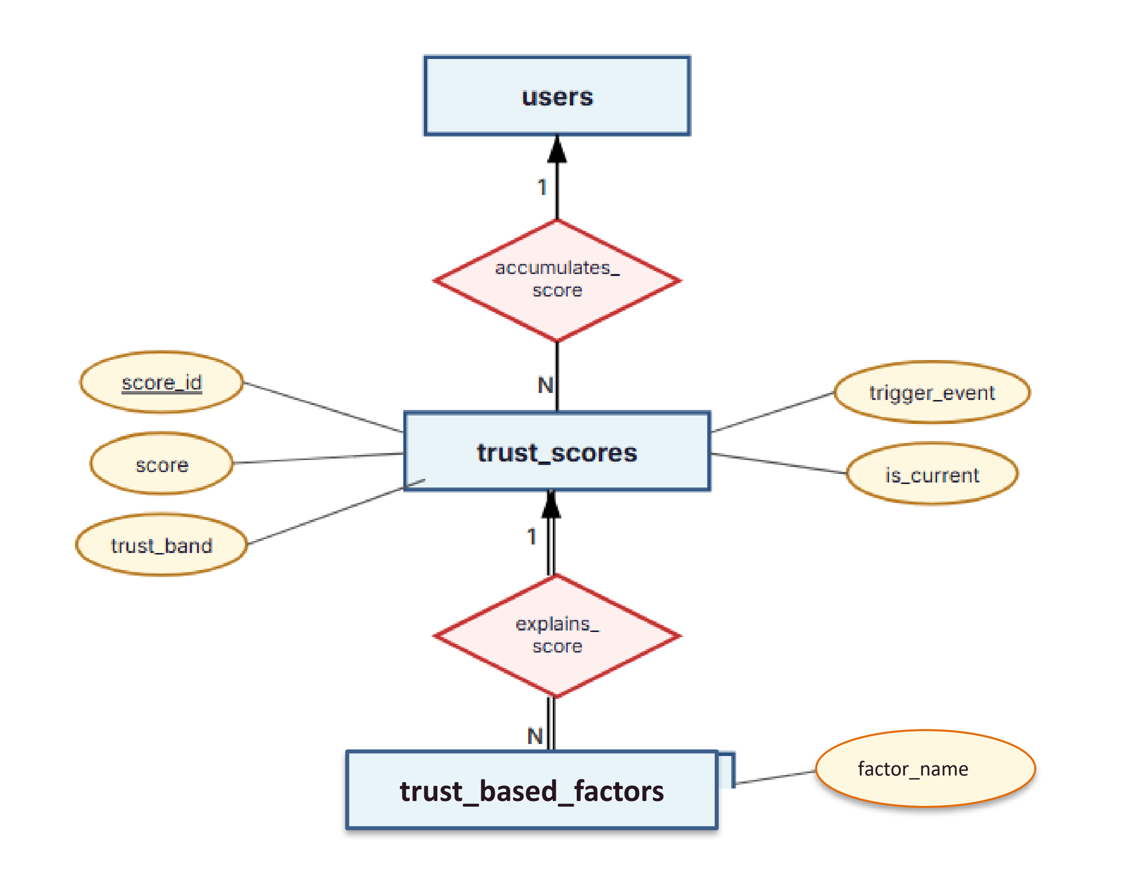
user_profiles, institutions, verification_requests, verification_documents

## Key Relationships

users → user_profiles (1:1) — separates identity from biography
user_profiles → institutions (N:1) — eliminates redundancy
users → verification_requests (1:N) — supports re-verification
verification_requests → documents (1:N) — multiple docs per request

# Module 3 — Trust Engine

Explainable, historical trust scoring — the platform's core innovation




## Purpose

Generates an explainable trust score for each borrower

Each recalculation creates a new immutable snapshot — history is never overwritten

Factors detail why a score was assigned (e.g., +20 for identity verified)

## Entities

trust_scores, trust_score_factors

## Key Relationships

users → trust_scores (1:N) — accumulates historical snapshots

trust_scores → factors (1:N, total–total) — every score must have factors; every factor belongs to a score

## Design Rationale

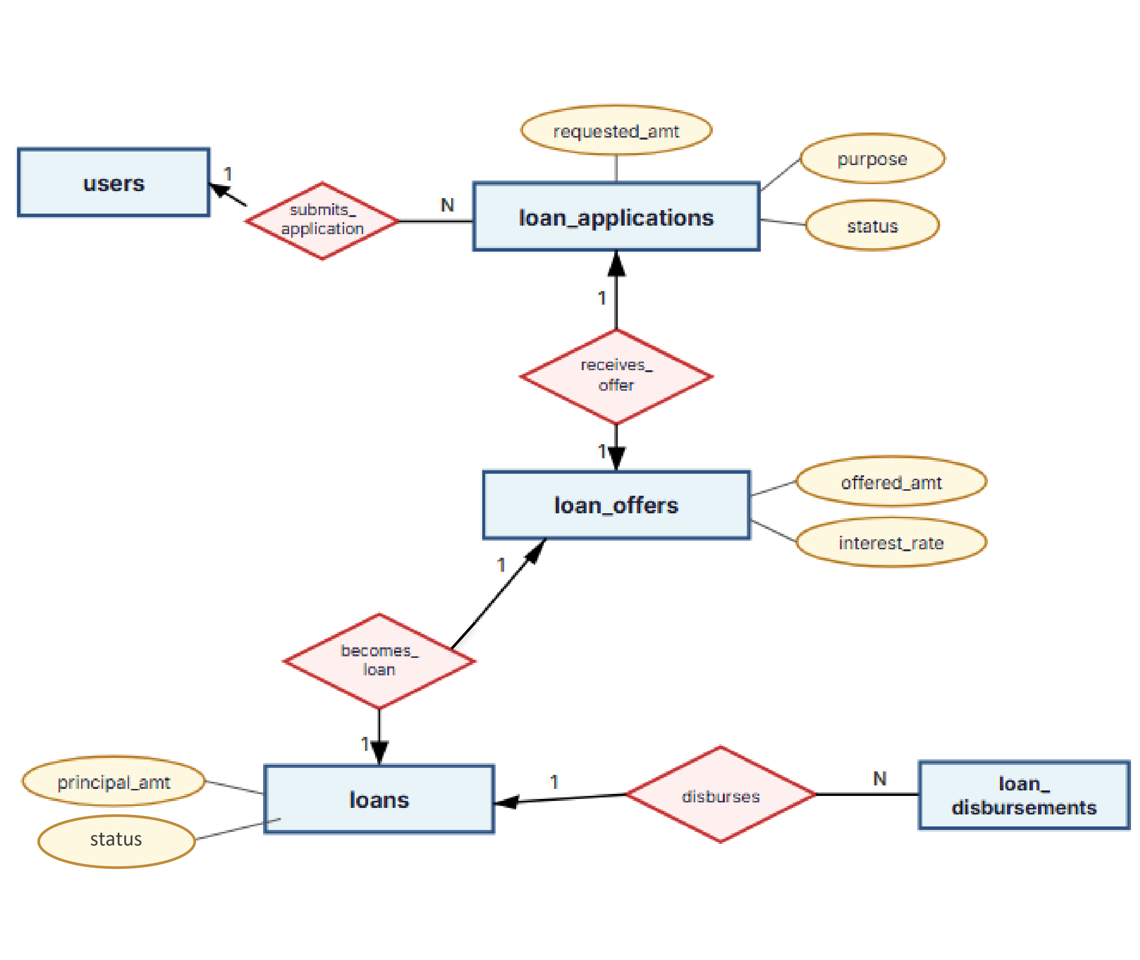
Evaluators must understand why a borrower scored 72, not just see "72"

is_current flag enables fast lookup without subqueries

Supports trend analysis, auditing, and fairness verification

# Module 4 — Loan Application & Loan Lifecycle

Application → Offer → Loan → Disbursement — four distinct lifecycle stages




## Purpose

Application, Offer, Loan, and Disbursement are four distinct entities representing different lifecycle stages

A request may never become a loan — rejected applications must persist

Partner may offer different terms than requested

## Entities

loan_applications, loan_offers, loans, loan_disbursements

## Key Relationships

application → offer (1:1) — partner counter-offer preserves original request

application → loan (1:1) — loan exists only after approval

loan → disbursements (1:N) — approval ≠ money transferred

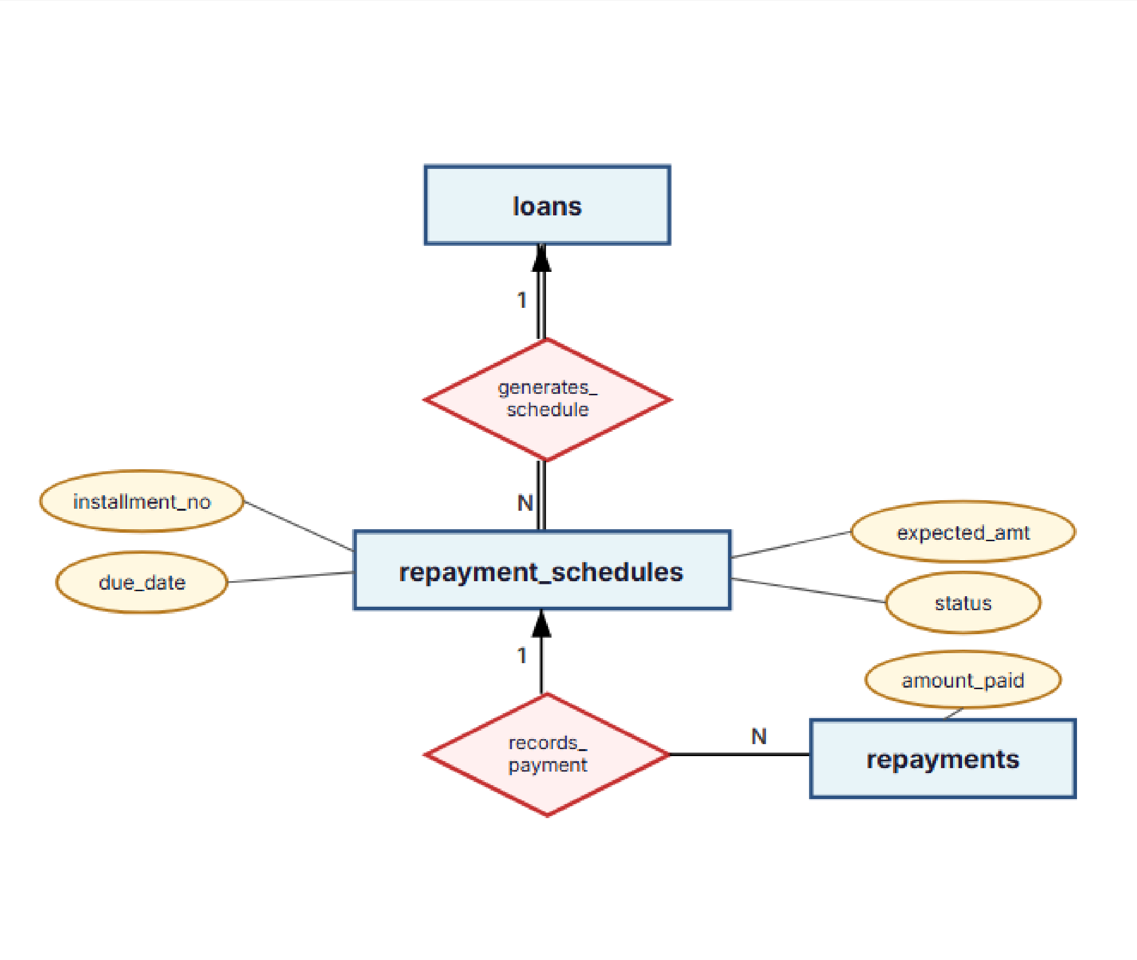
## Design Rationale

Mixing lifecycle stages would create NULL-heavy records and complex status logic

trust_score_id on application captures the score snapshot at submission time

# Module 5 — Repayment System

Expected installments vs. actual payments — never overwrite financial records




## Purpose

Expected installments (the plan) are separate from actual payments
Supports partial payments, failed attempts, and late payments
Financial records are append-only — never overwritten

## Entities

repayment_schedules, repayments

## Key Relationships

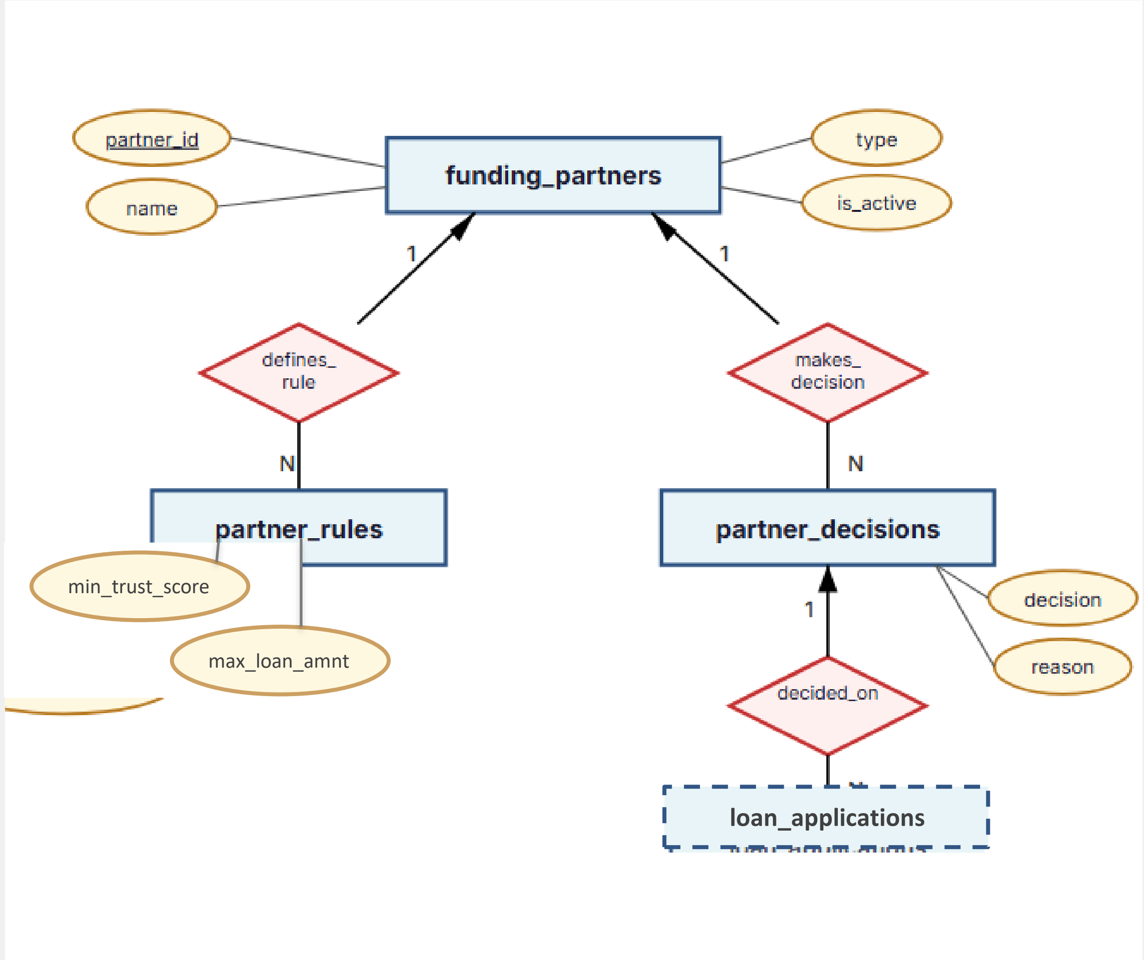
loans → repayment_schedules (1:N, total–total) — every loan generates a schedule
repayment_schedules → repayments (1:N) — multiple payments per installment

## Design Rationale

Enables overdue detection by comparing due_date with actual payments
Supports trust score recalculation triggers on repayment events
Outstanding balance is calculated, not stored (avoids derived-data issues)

# Module 6 — Partner Integration

ShohojRin is not a lender — partners are. Rules, decisions, and evaluations.




## Purpose

Partners (banks, NGOs, MFIs) define their own lending criteria as data, not code

Decisions are stored separately from applications — an application may receive multiple evaluations

## Entities

funding_partners, partner_rules, partner_decisions

## Key Relationships

funding_partners → partner_rules (1:N) — each partner has configurable criteria

funding_partners → partner_decisions (1:N) — decision history per partner

loan_applications → partner_decisions (1:N) — multiple partners may evaluate one application

## Design Rationale

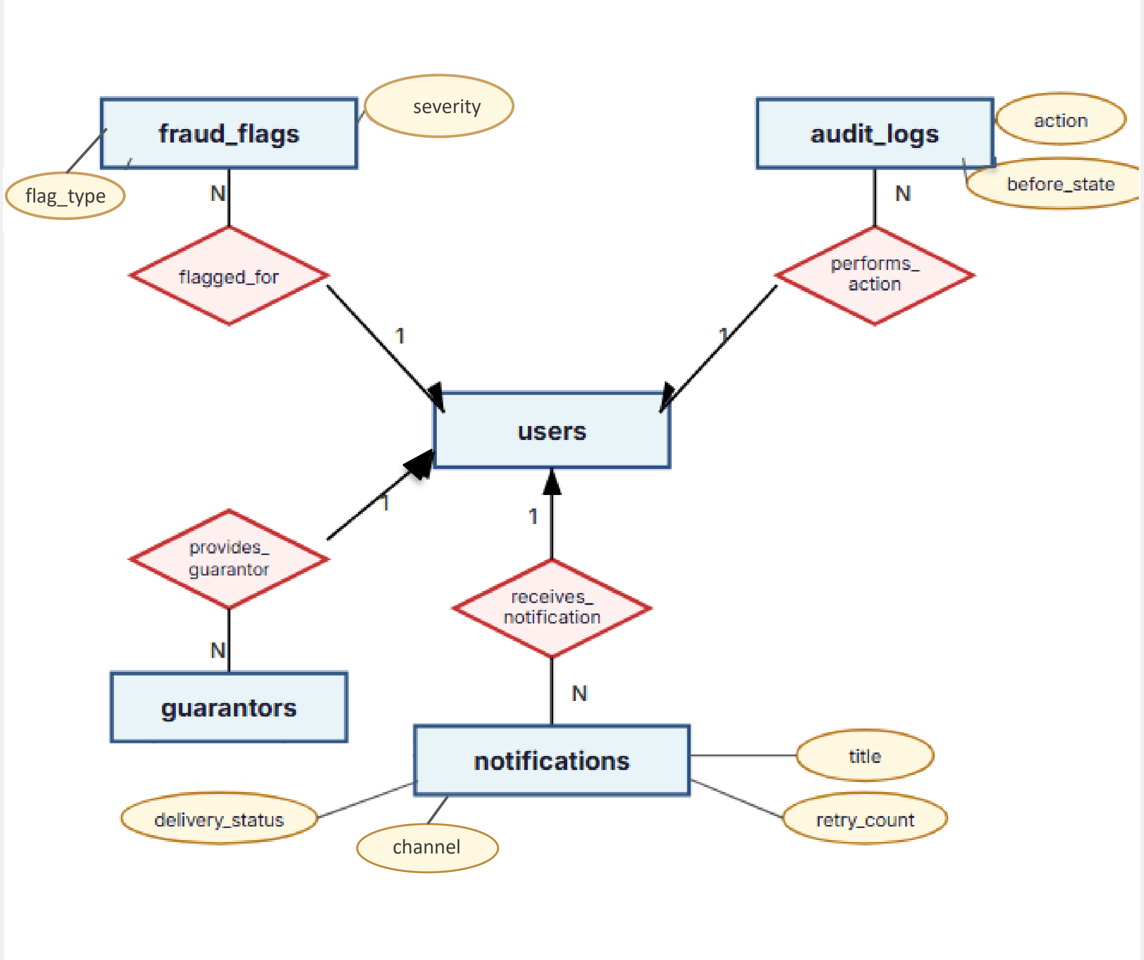
Adding a new partner requires no schema changes — only data insertion

Rules support effective-dating for versioning

Preserves full decision history for audit and partner analytics

# Module 7 — Fraud Detection, Audit Logs & Notifications

Cross-cutting concerns — security, compliance, and communication




## Purpose

Fraud flags — event-based suspicious activity records; never auto-deleted

Audit logs — immutable, append-only records of all sensitive operations

Notifications — multi-channel delivery with retry support

## Entities

fraud_flags, audit_logs, notifications, guarantors

## Key Relationships

users → fraud_flags (1:N) — user may be flagged multiple times

users → audit_logs (1:N) — tracks who, when, what, before/after

users → notifications (1:N) — supports email, SMS, in-app

users → guarantors (1:N) — borrower lists guarantor references

## Design Rationale

Audit logs use JSONB for before/after state — flexible across all entity types

Fraud flags support admin review workflow with severity levels

All three tables are cross-cutting — they reference users but serve independent concerns

# System Workflow — End-to-End Process

From registration to repayment — how the complete ShohojRin system operates